Code-Layout, Readability and Reusability

Date	**07/07/2026**
Team ID	SWTID-2026-5714
Project Name	OptiCrop: Smart Agriculture Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Wellorganized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Code indentation is consistent across all files	Yes	Consistent indentation maintained throughout app.py, the Jupyter notebook, and HTML templates
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized with app.py, crop_model.pkl, scaler.pkl, requirements.txt, data/, templates/, and OptiCrop.ipynb
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	temperature, rainfall, humidity, ph, N, P, K
4	Function / Method Names	Functions are descriptively named	Yes	Functions used for model loading, input validation, and crop prediction
5	Code Comments	Inline and block comments explain logic	Partial	Key ML and prediction functions are commented
6	Modular Design	Code is split into reusable functions/modules	Yes	Frontend templates, ML model, dataset preprocessing, and prediction logic are organized separately
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code is avoided by reusing functions and keeping the code clean

8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic input validation and error handling implemented in the /predict route

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High/Medium/Low)
1	Crop prediction model	Python, scikitlearn	Crop prediction module	High
2	Data preprocessing module	Python, Pandas	Model training & prediction	High
3	Flask API	Python, Flask	Web application backend	High
4	HTML templates (Home / About / Predict Crop / Result)	HTML, CSS	Farmer-facing web pages	Medium
5	Dataset loader	Python, Pandas	Training and testing process	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Project is well organized with separate files for the ML model, dataset, application, and documentation.
Readability	5	Code uses meaningful variable names, proper indentation, and clear function names.
Reusability	5	Reusable modules like the crop prediction model, dataset loader, and preprocessing/scaling are used across the project.
Documentation / Comments	4	Basic comments and project documentation are provided, but more detailed inline documentation can be added.

Overall Score	5	The project follows good coding standards, modular design, and reusable components with a clean structure.